Examiner only

8. This question is about momentum.

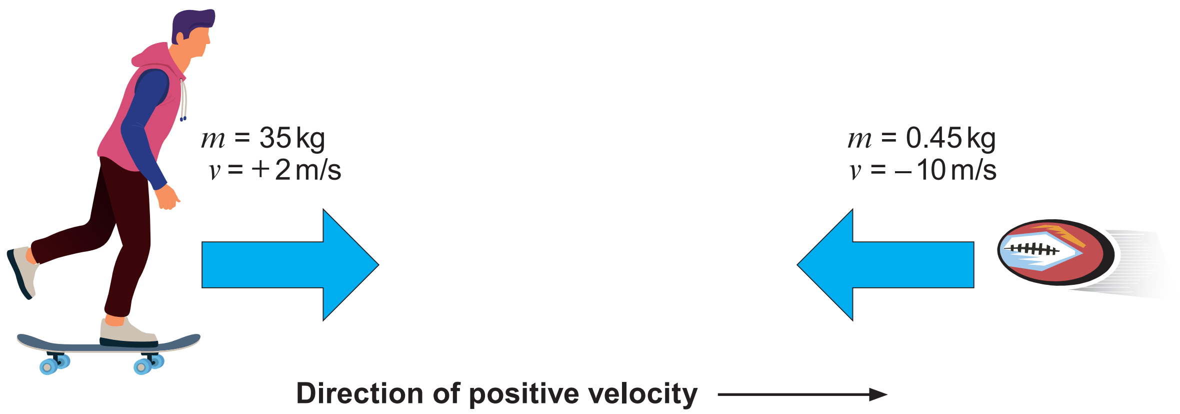
(a) State the principle of conservation of momentum. [2]

....

....

....

(b) A skateboarder of mass 35 kg moving with a velocity of 2 m/s catches and holds on to a 0.45 kg rugby ball thrown directly towards him at 10 m/s.



Use the equation:

$$\text{momentum} = \text{mass} \times \text{velocity}$$

to calculate the **total** momentum of the skateboarder and the ball before the collision and give the unit. [4]

Total momentum = ...

Unit = ...

 (3420UB0-1)

Examiner only

(c) Use an equation from page 2 to calculate the velocity of the skateboarder after he catches the rugby ball. [2]

Velocity = ... m/s

(d) Tomos suggests that **kinetic energy** is conserved in this collision.
Explain, by using calculations, whether or not you agree.
Use an equation from page 2. [4]

....

....

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END OF PAPER